Algebra SEP Worksheet 10 8/18/22

Miscellaneous/Review

Arcana

∗ **1.**

On this problem, no justification is required; only the answer will be graded.

(a) Give an example of a non-diagonalizable matrix in $\mathrm{GL}_3(\mathbb{F}_p)$ (for some p).

(b) Give an example of a nontrivial proper normal subgroup of $\mathrm{GL}_3(\mathbb{F}_p)$ (for some p).

(c) Give an example of a one-sided ideal (either left or right) in $M_2(\mathbb{C})$ which is not a two-sided ideal.

(d) Give an example of a $\mathbb{Z}$-module which is flat but not free.

(e) Give an example of an exact sequence of abelian groups which is not split.

∗ **2.**

On this problem, no justification is required; only the answer will be graded.

(a) Give an example of a finite extension of $\mathbb{Q}$ which is not Galois.

(b) Give an example of a non-trivial, proper, two-sided ideal in $M_2(\mathbb{Z})$.

(c) Give an example of an injective map $M \to N$ of $k[x]$-modules (k a field) and another $k[x]$-module R such that $M \otimes R \to N \otimes R$ is not injective.

(d) Give an example of a commutative ring R and prime ideals p, q, q' in R such that the extension $\overline{q}$ of q to the localization R_p is prime, while the extension $\overline{q'}$ of q' to R_p is not prime.

∗ **3.**

On this problem, no justification is required; only the answer will be graded. Let S_3 be the symmetric group on 3 letters, and let R be the group ring $\mathbb{Z}[S_3]$.

(a) Write down a nonzero element of R which is a zero-divisor.

(b) Write down an element in the center of R which is not in $\mathbb{Z}$.

(c) Write down an R-module which is not a free module.

∗ **4.**

Decide if the following statements are true. If so, provide a proof, if not, provide a counterexample. Let M, N be finitely generated modules over a commutative ring A. Let $f\colon M \to N$ be a homomorphism of A-modules, and let $g\colon B \to A$ be a homomorphism of commutative rings.

(a) “If f is injective, so is the induced morphism $M \otimes_A M \to N \otimes_A N$.”

(b) “If f is surjective, so is the induced morphism $M \otimes_A M \to N \otimes_A N$.”

(c) "If g is injective, so is the induced morphism $M \otimes_B M \to M \otimes_A M$."

Scattered Esoterica

$*$ **5.**

Let $K \subseteq F \subseteq E$ be fields with $E = F[\alpha]$ and with $\alpha^n \in F$ for some positive integer n.Suppose K contains a primitive nth root of unity, and let L be a field with $K \subseteq L \subseteq E$ and $L \cap F = K$.

(a) If L is Galois over K, show that $L = K[\beta]$ for some element β with $\beta^n \in K$.

(b) Show by example that L need not be Galois over K. (Hint. Take $n = 2$.)

$*$ **6.**

Consider the field with 11 elements $F = \mathbb{F}_{11}$. Let G denote the cyclic group of order 11 (written multiplicatively), with generator $r \in G$. Denote by FG the group algebra of G (also sometimes denoted $F[G]$). We consider r as an element of FG, and let $T\colon FG \to FG$ be the F-linear map given by $T(x) = rx$ for all $x \in FG$. Find the Jordan canonical form of T.

$*$ **7.**

Let F be a field of prime characteristic p. Suppose $E = F(\alpha)$ is a simple extension such that $\alpha \notin F$ but $\alpha^p - \alpha \in F$.

(a) Find $[E : F]$.

(b) Prove that E/F is a Galois extension.

(c) Find the Galois group $\mathrm{Gal}(E/F)$.

Hint: What is $(x + 1)^p - (x + 1)$ in characteristic p?

$*$ **8.**

Let k be a field, and let $R = M_2(k)$, the (noncommutative) ring of 2×2 matrices over k.

(a) Give an example of a simple left R-module M, and a simple right R-module N.

(b) Can you find another simple left R-module M' which is not isomorphic to M? (Either find one, or explain why there is none.)

(c) Compute $\dim_k N \otimes_R M$.

$*$ **9.**

Let V be a non-zero finite dimensional vector space over $\mathbb{C}$. Given a linear transformation $A\colon V \to V$, show that the following are equivalent.

(i) There exists a linear transformation $P\colon V \to V$ such that $P^2 = I$ and $AP = -PA$.

(ii) There exists an invertible linear transformation $P\colon V \to V$ such that $AP = -PA$.

(iii) There exists a direct sum decomposition $V = V_1 \oplus V_2$ such that $AV_1 \subset V_2$ and $AV_2 \subset V_1$.

(Hint: Thinking about (iii) $\Rightarrow$ (i) helped me to think about (ii) $\Rightarrow$ (iii).)